

Paper 25 - Energetic Traversal Maps

CQE-paper-25

CQECMPLX Formal Paper Series

Review-facing PDF built 2026-06-12

Construction Basis

Add energy and traversal ledgers to cross-language, figure, material, and fold transformations.

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- promoted formal paper: production\formal-papers\CQE-paper-25\FORMAL_PAPER.md
- paper kernel manifest: production\paper-kernels\papers\CQE-paper-25\paper_kernel_manifest.json
- verifier: production\formal-papers\CQE-paper-25\verify_energetic_traversal.py
- receipt: production\formal-papers\CQE-paper-25\energetic_traversal_receipt.json (CQE-paper-25: pass_with_open_obligations)
- body: production\papers\CQE-paper-25\PAPER-BODY.md
- source: production\papers\CQE-paper-25\SOURCE.md
- formal: production\papers\CQE-paper-25\01-CQE-formal\FORMAL.md
- tool: production\papers\CQE-paper-25\02-CQE-tool\TOOL.md
- workbook: production\papers\CQE-paper-25\03-CQE-workbook\WORKBOOK.md
- recovered_output: production\lib-forge\recovered\papers_output\CQE-paper-25.md

Paper 25 - Energetic Traversal Maps

Abstract

Paper 25 introduces the energetic traversal map as a unit-agnostic accounting kernel for CQE transports. A transport may move a registered state from one formal surface to another only when it emits a replayable Noether-Shannon- Landauer boundary row. The row computes $\theta = \alpha * N + \beta * S + \gamma * L - \text{absorption}$, marks whether the step closes internally, and carries any residue as an obligation instead of erasing it.

The closed result is not physical energy. It is an accountable traversal receipt. The verifier proves that the live package computes the NSL boundary term, that path totals add from step totals, that the four-layer transport spine is present with its open lifts visible, and that the default analog state cost is the verified $2 + 6$ VOA split $Z(q) = 2q^0 + 6q^5$. A physical joule claim, least-action claim, thermodynamic optimum, or unification of Noether, Shannon, and Landauer law requires a later calibrated domain proof.

Closed Evidence

The formal receipt is generated by ``production/formal-papers/CQE-paper-25/verify_energetic_traversal.py``. It imports ``NSLTerm`` from ``lattice_forge.ledger.nsl``, ``verify_transport_obligations`` from ``lattice_forge.transport_obligations``, and ``verify_voa_sector_decomposition`` from ``lattice_forge.centroid_voa``.

The receipt status is ``pass_with_open_obligations``. The transport spine has four rows. Two rows are demonstrated locally: ``LCR_TO_D4_AXIS_SHEET`` and ``D4_TO_J3O_DIAGONAL_CARRIER``. Two rows remain bounded or registered lifts: ``J3O_TO_G2_F4_T5A_ROUTE`` and ``EXCEPTIONAL_ROUTE_TO_NIEMEIER_LANDING_FORMS``. Their presence is not hidden; the receipt reports them as open lifts.

The VOA default analog cost verifier passes with weight distribution ``{0: 2, 5: 6}`` and partition $Z(q) = 2q^0 + 6q^5$. In the worked traversal, the closed normalized path has three steps and $\theta_{\text{path}} = -0.35$. The open path has two steps, includes an uncalibrated domain row, and reports $\theta_{\text{path}} = 0.8999999999999999$. Both open rows are carried as obligations.

Definitions

An active transport is the current move from a source object to a target object. Its center ``C`` is the preserved invariant the move claims to carry. Its opposed readouts ``L`` and ``R`` record what is preserved and what remains as boundary residue.

An NSL boundary term is the tuple ``(N, S, L, absorption, alpha, beta, gamma)``. ``N`` is the conservation mismatch, ``S`` is the information mismatch, ``L`` is the irreversible execution cost, and ``absorption`` is the declared capacity that may absorb the mismatch. The closed step gate is $\theta \leq 0$.

An energetic traversal map is an ordered path of transport rows. Its path cost is the sum of the step costs after the rows have been normalized into compatible accounting units. A traversal is closed only when the path total closes and no step is marked uncalibrated.

A unit policy is part of the receipt. The default paper policy is ``normalized_analog_units``; a

physical-energy reading is invalid until a domain calibrates the NSL terms into physical units.

Claims

- Every accepted CQE traversal can carry a replayable NSL row per step.
- The step gate is exactly the package gate:
 $\text{`theta} = \alpha * N + \beta * S + \gamma * L - \text{absorption}`$, with closure iff $\text{`theta} \leq 0`$.
- For normalized rows, traversal totals are additive:
 $\text{`theta\_path} = \text{sum}(\text{theta\_i})`$.
- The package's four-layer transport-obligation spine is the canonical traversal surface for this paper, and its open lifts remain visible.
- The verified VOA sector split supplies a default analog cost: two vacuum states carry weight 0 and six excited states carry weight 5.
- A traversal receipt does not become a physical-energy proof unless a later domain calibration proves the required unit map.

Theorem 25

An energetic traversal is valid in the CQE kernel exactly when it emits a replayable NSL boundary row for each step, sums those rows into a path total, marks closure or obligation without deletion, and declares whether its units are analog-normalized or physically calibrated.

Proof

``NSLTerm`` is a frozen data object whose ``theta`` property evaluates $\alpha * N + \beta * S + \gamma * L - \text{absorption}$. Its ``closes_internally`` property is the Boolean $\text{`theta} \leq 0`$. The verifier checks a weighted sample with ``sample_theta = 2.4`` and confirms that a positive term does not close while a negative term does.

The verifier then calls ``verify_transport_obligations``. The returned spine has four rows, all required fields present, and passing local witnesses. It reports ``pass_with_open_lifts``, not a total closure. This is the needed boundary behavior: demonstrated rows can be used directly, while bounded and registered rows are still carried as named obligations.

For traversal additivity, the verifier constructs two path receipts. Each path stores per-step NSL dictionaries, per-step ``theta``, the unit policy, and the path total. The receipt recomputes each ``theta_path`` as the sum of its step rows. The normalized path closes with ``theta_path = -0.35``. The second path has positive boundary residue and an uncalibrated domain unit policy, so it remains open with ``theta_path = 0.8999999999999999``. No row is discarded.

Finally, the verifier calls ``verify_voa_sector_decomposition``, which passes with $Z(q) = 2q^0 + 6q^5$. This gives a package-native analog weight to chart states before any domain-specific physical calibration is attempted. Therefore the energetic traversal map is closed as an accounting kernel and open as a physical-energy model.

Worked Example

The closing path moves from $\{0,1\}^3$ into a D4 axis/sheet token, then into a ``J3(O)`` diagonal carrier, then into the bounded exceptional route. Its states include two vacuum-weight rows and one excited-weight row. Every row uses ``normalized_analog_units``, every step closes, and the

path sum is negative.

The open path begins with the same chart-to-D4 surface and then attempts a registered Niemeier landing-form lift. Its first row has positive boundary residue. Its second row also has positive residue and is marked `uncalibrated\_domain\_units`. The path therefore becomes a useful diagnostic receipt, not a passed claim.

Open Obligations

Physical unit calibration remains open. The paper does not convert NSL rows to joules.

Absorption measurement remains open. `absorption\_capacity` must be supplied by a domain measurement or declared policy.

Least-action, geodesic, or thermodynamic-optimum readings are not claimed. A smaller analog `theta\_path` is leaner only inside the receipt ledger.

Noether/Shannon/Landauer unification is not claimed. The paper uses those terms as named accounting components and inspiration anchors, not as a replacement for the underlying laws.

Suite Role

Paper 25 gives Papers 22-24 and later applied papers a shared traversal ledger. Material transforms, folding routes, board automata, CAD routes, market routes, and kernel handshakes can now expose the same question: did the requested transport close, where did it fail, and which residue must be carried forward?